

Experiment Runner Framework – design + UML class diagram

2026-06-04 (Aiko / Csaba)

Scope

Replace 221 launcher scripts in `signedkan_wip/experiments/` with one runner framework + YAML configs. CLAUDE.md §6.5 #3 has mandated this since 2026-05-11; this is the implementation.

The full design narrative is in `runner_architecture.md` (in the same directory). This LaTeX wrapper carries the formal UML class diagram below.

Class diagram

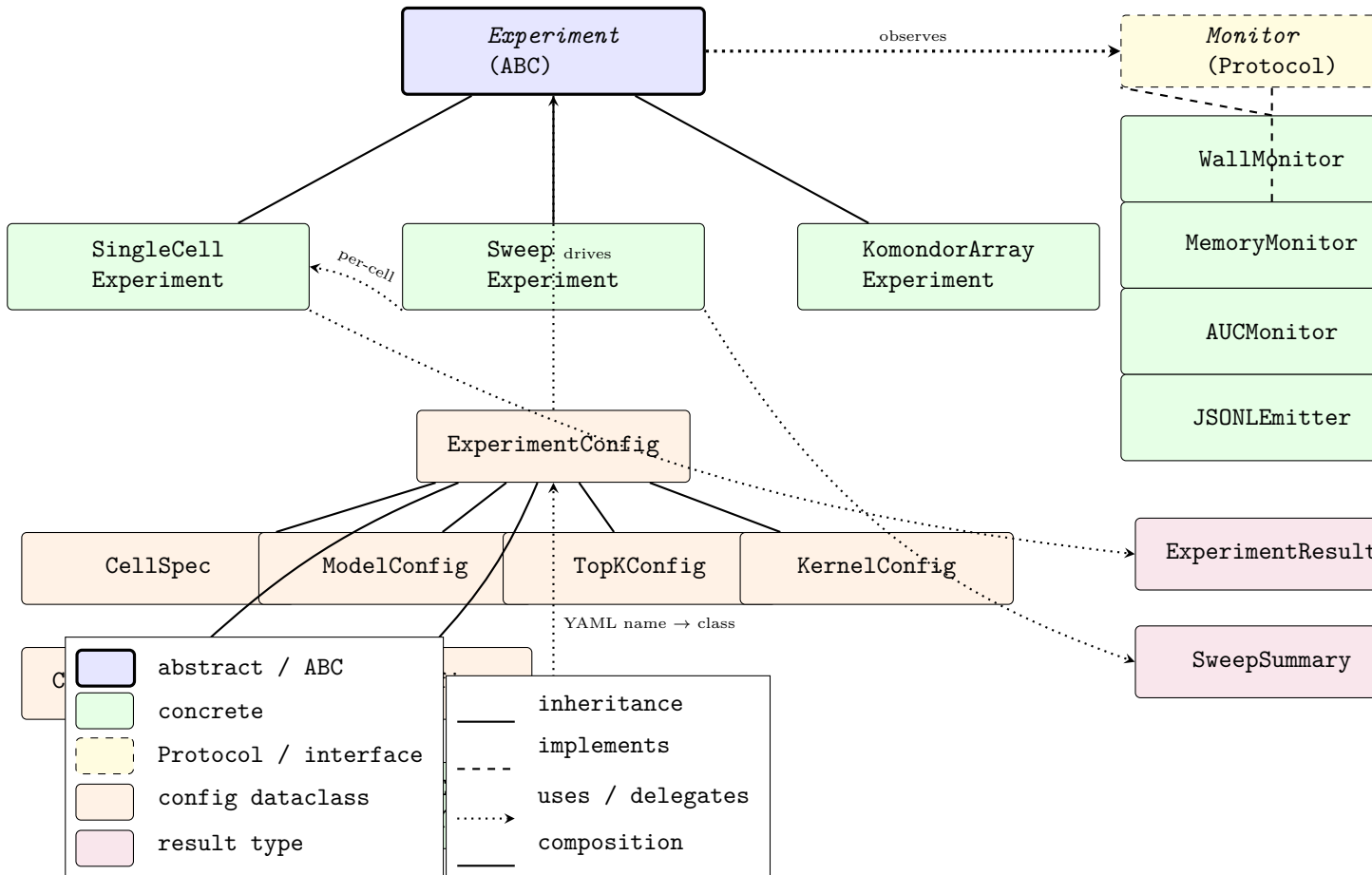


Figure 1: Experiment runner framework. *Italic* = abstract / protocol. Concrete subclasses fan out to the three deployment targets (single-cell, multi-seed sweep, Komondor array). `ExperimentConfig` is a composition of existing `runtime_config.py` dataclasses (`TopKConfig`, `KernelConfig`, ...), extended with cell / model / output specifications. Monitors are an Observer pattern attached to every `Experiment`; the registry maps YAML `experiment_class` names to concrete `Experiment` subclasses.

Mermaid version

A browsable Mermaid version of the same diagram is at `runner_class_diagram.mmd` (compatible with GitHub rendering and the user's IDE).